



Polystyrene brushes/TiO₂ nanoparticles prepared via SI-ATRP on polypropylene and its superhydrophobicity

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Abstract

Surface initiated-atom transfer radical polymerization (SI-ATRP) is a powerful tool for careful control of the materials surface properties, such as their wettability. In this study, superhydrophobic self-cleaning propylene (PP) surfaces were obtained by dipping the PP sticks into xylene at 125 °C containing dispersed hybrid core-brush TiO₂ nanoparticles (NPs). The TiO₂ NPs were functionalized with (3-chloropropyl) triethoxysilane (CLE) and they were employed for styrene grafting polymerization using CLE as an SI-ATRP initiator. DRIFT, TEM, SEM, WCA, optical profilometry, TGA and SAXS techniques were used for characterization. TEM images of the (NPs-CLE^a)-g-PS revealed that the shell brush sizes were (10.4 ± 0.5) nm, (11.5 ± 0.5) nm and (18.8 ± 0.5) nm for *a* (CLE/xylene ratio), equal to 5/95, 10/90 and 20/80 respectively; indicating the dependence of the polymer grafting on the initiator ratio. The hierarchical micro-nanostructure generated on PP sticks, the WCA and hysteresis also depended on the *a* used. The WCA and hysteresis reached values of (161 ± 7)° and (3.7 ± 0.4)°, respectively for *a* = 20/80. The present methodology employs a simple and cost-effective technique to produce durable superhydrophobic surfaces on PP, offering the possibility to be used in different polymeric substrates. Additionally, it can be easily scale up and integrated into industrial processes.

Keywords Core-brush nanoparticles · Surface initiated-atom transfer radical polymerization (SI-ATRP) · Modified surfaces · Superhydrophobicity · Self-cleaning

Introduction

The control of the wettability properties of materials is of great technological importance and has recently attracted the interest of both academia and industry [1, 2]. The field of

artificial superhydrophobic surfaces has been developed by mimicking natural objects [3, 4], such as the self-cleaning lotus leaf, the hydrophobicity of red rose petals, anisotropic wetting behavior on the rice leaf, hydrophobicity of shark skin in the interface liquid/solid, gecko toe, insect wings, etc. [5]. All of these examples are related to the unique micro-nano hierarchical structures that are governed by both the chemical composition and the topographical organization [6–8]. By being able to control wetting behavior of a liquid on solid surfaces many applications became feasible in various fields, such as in micro-fluidics, the separation of oil/water dispersions, self-cleaning, drag reduction, anti-fogging, anti-bacteria, anti-fouling, anti-icing, and corrosion resistance, among others [9–11].

Many physical and chemical methods have been proposed to manufacture superhydrophobic surfaces based on two basic rules that include materials with low surface energy and optimal organization of surface micro-nanostructures [12, 13]. The production of artificial superhydrophobic surfaces can be achieved by lithography, template based techniques, chemical treatments [14–18], plasma treatments,

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self-assembly, layer-by-layer (LBL) deposition [19, 20], colloidal assembly phase separation, electrospinning, among others [21–24].

Among chemical methods, surface grafting is, in fact, one of the most widely used approaches to modify material surface properties [25], in which grafting covalently links specific macromolecules to the polymer surface, thus enabling a precise tailoring of the material surface properties [26, 27]. The advantages associated with this technique include: the long-term stability of the covalently grafted layer; the ability to place macromolecules at specific locations on the material surface with high densities, and the capacity to incorporate multiple types of molecules [28–30].

The preparation of polymer brushes via surface initiated-atom transfer radical polymerization (SI-ATRP) allows precise control of surface chemical composition and its hierarchical nano-microstructure [31, 32]. Those properties of SI-ATRP methodology have attracted increasing attention for the fabrication of artificial superhydrophobic materials. For example, cotton [33] and filter paper [34] with simultaneous superhydrophobic and superoleophilic properties were prepared for oil/water separation by polymer grafting via SI-ATRP. This technique was also employed to fabricate antifouling superhydrophobic surfaces of poly(ethylene terephthalate) fabrics [35]. Poly (*N*-isopropyl acrylamide) coatings on glass surfaces by SI-ATRP were also efficiently produced [36]. The film properties allowed the release of marine microorganisms attached to the surface depending on the seawater temperature change, the grafting density and the thickness of the polymeric layer. The addition of nanoparticles (NPs) functionalized previously or in situ has provided a new kind of hybrid superhydrophobic materials. For instance, superhydrophobic fluorinated polymer brush grafted onto nanostructured SiO₂ NPs by SI-ATRP have been used to repel efficiently proteins, producing a material with antifouling and self-healing properties [37]. In another example, a biomimetic superhydrophobic SiO₂ NPs/PDMS film was prepared to simulate the shark skin surface by SI-ATRP of poly(2-perfluorooctylethyl methacrylate), presenting also self-healing and drag reduction properties [29].

We have prepared in the past superhydrophobic propylene (PP) surfaces by the use of functionalized NPs [16]. In particular the use of TiO₂ NPs was interesting due to its photocatalytic properties that can lead to the reversible switching between superhydrophobicity and superhydrophilicity [38]. Recently, superhydrophobic PP surfaces with water contact angles of over 158 degrees have been prepared by SI-ATRP using the dendron acryl Behera's amine [18]. The dendritic polymer was successfully grafted on the TiO₂ NPs surfaces and specific wettability properties on the PP surfaces were obtained. High water adhesion or self-cleaning conditions were measured by control of the organic silane concentration and in consequence the

SI-ATRP process. In addition, the dendritic polymer covalent bound to TiO₂ NPs permitted to obtain more stable superhydrophobic properties over time and good adhesive properties. This experience allowed us to provide interesting contributions to the structure/property relationship of the surface topology and the influence of nature and size of the building blocks selected to create superhydrophobicity. In spite we have reported a novel synthesis procedure for the construction of hybrid superhydrophobic nanomaterials from dendronized TiO₂ NPs, the high cost of the dendron used may not allow a scale up of the process. Therefore the simple dip-coating process has to be coupled with the use of low cost chemical components. The actual challenge in a final application is to decrease the cost of each step of the process keeping its properties. In the present study, we report an easy and simple methodology for preparing superhydrophobic PP surfaces with super-repellent and self-cleaning properties. Thereby, in a first step, core-brush TiO₂ NPs prepared by in situ grafting were obtained by using SI-ATRP and a low cost industrial monomer, such as styrene. Then, these core-brush structures were used as building blocks to modify the PP surfaces by dip-coating. The method uses a simple technique and it is cost-effective. In addition, it offers the possibility to produce superhydrophobic surfaces on different polymeric substrates and it can be easily integrated into industrial processes.

Materials and methods

Materials

The polypropylene (PP) sticks were injection-molded about 30×10 mm in size and of 3 mm thickness. Titanium dioxide nanoparticles (AEROXIDE® TiO₂-P25) were acquired from the Degussa Corporation. Styrene, (3-chloropropyl) triethoxysilane (CLE), hexamethyl triethylenetetramine (HMTETA), CuCl and CuCl₂ were purchased and used without treatment from Sigma Aldrich. Xylene was obtained from Merk. Polystyrene (PS) pellets were obtained from Aldrich Chemicals Co., UK.

Surface modification of TiO₂ NPs

The TiO₂ NPs surface modification with polystyrene brushes was carried out by SI-ATRP divided in two steps. First, TiO₂ NPs functionalization was achieved by silanization process with an organic silane agent, (3-chloropropyl) triethoxysilane, CLE. Subsequently, the functionalized NPs were employed in a second step, for styrene grafting polymerization using CLE as an SI-ATRP initiator.

Functionalization of TiO₂ nanoparticles for SI-ATRP reaction

The functionalization of TiO₂ NPs was achieved by previously reported silanization process with two organic silane agents, (3-chloropropyl) triethoxysilane (CLE) [18]. Briefly, 50 mg of the pure NPs TiO₂ were dispersed into 5 mL of organosilane/xylene solution and the mixture was sonicated for 3 h. The organosilane presence onto the TiO₂ NPs surfaces allowed the polymerization grafting to be initiated via SI-ATRP. Three concentrations ratios of organic silane/xylene were used ($a = 5/95$, $10/90$ and $20/80$) in order to evaluate the initiator influence in the polymerization on the TiO₂ NPs.

Polystyrene SI-ATRP grafting on functionalized nanoparticles

Polystyrene (PS) grafting was performed by SI-ATRP on previously functionalized NPs. A portion of 50 mg of the functionalized NPs were dispersed in 10 mL of xylene into a two-necked flask equipped with a nitrogen inlet, after which, the styrene monomer and the HMTETA ligand were added. Three freeze–pump–thaw cycles were performed in order to de-gas the system. Then, a catalyst system (CuCl, CuCl₂) was added under nitrogen flux. Both the ligand and the catalyst system were used at the same proportion as the NPs, while styrene was used in excess ($1.92 \cdot 10^{-3}$ mol.mg NPs⁻¹). The SI-ATRP reaction was carried out at 90 °C during 18 hs with the polymerization being stopped by opening the flask to permit the entry of air. The core-brush NPs with brushes of PS were purified and recovered by five redispersion–centrifugation cycles in dichloride methane to eliminate the non-attached chemicals (7000 rpm, 15 min). The product obtained were the serie (NPs-CLE^a)-g-PS with $a = 5/95$, $10/90$ and $20/80$, see Table 1.

Construction of superhydrophobic surfaces

The superhydrophobic surfaces were prepared by following a dip-coating procedure previously reported by our group [16]. Concisely, core-brush NPs (50 mg) were redispersed in xylene at 125 °C and a PP stick was submerged for 3 s. Then, the modified stick was dried in an oven at 100 °C.

Characterization techniques

Core-brush TiO₂ NPs characterization

Infrared spectroscopy Infrared spectra were acquired using a Nicolet 360 spectrometer with a Diffuse Reflection Fourier Transform (DRIFT) data a Smart DRIFT accessory, 32 scans, resolution of 4 cm⁻¹. Samples were prepared by mixing the NPs with KBr until a final composition of 3% w/w and dried at 35 °C overnight. The resulting data were processed using Omnic 8.0 software.

Transmission electron microscopy TEM images of the pure and core-brush TiO₂ NPs were recorded employing a JEOL 1200 EXII electronic microscope operated at 80 kV. Samples were prepared by drop-casting onto standard carbon-coated colodion film copper grids (200-mesh). For each sample, a minimum of four different areas were evaluated and the estimated size of the NPs was calculated using the Image J public domain processing software.

Thermogravimetric analysis TGA data were obtained using sealed aluminium pans inside DTG 60 Shimadzu equipment, heated up from room temperature to 600 °C ($10\text{ }^{\circ}\text{C} \cdot \text{min}^{-1}$ under nitrogen flux $50\text{ mL} \cdot \text{min}^{-1}$).

In addition, the estimated number of surface grafted groups per area unit (N_{exp}) and percentage of grafting (%G) of modified TiO₂ NPs by polystyrene brushes was calculated using TGA data according to Eq. 1 and 2 respectively. Where, m_{org} is the final mass difference, m_{inorg} is the remaining inorganic mass, S_{NP} , V_{NP} and ρ correspond to the surface, volume and density of the TiO₂ NPs from geometric estimations, N_A is the Avogadro's number, M_{org} is the molecular weight of the incorporated organic group and m_0 is initial mass. For the nanoparticle density, ρ , we assumed a value of 4.26 g/cm^3 from commercial specifications.

$$N_{\text{exp}} = \frac{\left(\frac{m_{\text{org}}}{m_{\text{inorg}}}\right) \rho V_{\text{NP}} N_A}{M_{\text{org}} S_{\text{NP}}} \quad (1)$$

Table 1 Different syntheses realized and core-brush hybrid materials produced from PS ($1.92 \cdot 10^{-3}$ mol.mg NPs⁻¹) grafting by SI-ATRP carried out at 90 °C during 18 h

Series	Organic silane agent	Organic Silane/Xylene concentration (v/v)	Core-brush material
(NPs-CLE)-g-PS	CLE	5/95	(NPs-CLE ^{5/95})-g-PS
		10/90	(NPs-CLE ^{10/90})-g-PS
		20/80	(NPs-CLE ^{20/80})-g-PS

$$\%G = 100 \frac{m_{org}}{m_o} \quad (2)$$

Small-angle X-ray Scattering SAXS experiments for pure and core-brush TiO₂ NPs were realized using a Xeuss 1.0 instrument. The pure or core-brush TiO₂ NPs were re-dispersed in ethanol and the measurement conditions set were: sample-to-detector distance 537 mm and the x-ray wavelength 0.18 nm. All experiments were performed at 22 °C constant temperature, with an optical path sample chamber of 0.5 mm². The scattering curves were plotted as functions of absolute intensity, *I*, versus *Q*, and the data was processed using the Guinier approximation [39].

Modified PP surface characterization

Scanning electron microscope SEM images of modified PP surfaces were recorded employing an EVO 50 ZEISS and FE-SEM Sigma microscope with electron acceleration between 3 and 5 kV. All PP samples were previously metalized with gold.

Profilometer measurements The roughness profile and the RMS (root mean squared) roughness of unmodified and modified PP surfaces was obtained by using the optical profilometer Bruker Centaur GTK. For each unmodified or modified PP sample, three different sticks were studied, with all of these measurements being repeated at least five times at different positions.

Water contact angle measurements The static water contact angle (WCA) and contact angle hysteresis (CAH) were measured at ambient temperature using a drop of deionized water (4 µL), which was gently deposited on the substrate by means of a micro syringe. For CAH evaluation the advancing and receding contact angles were measured at the front

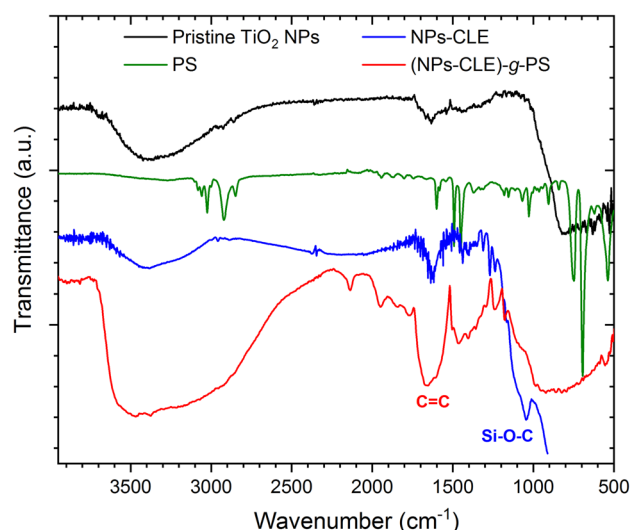
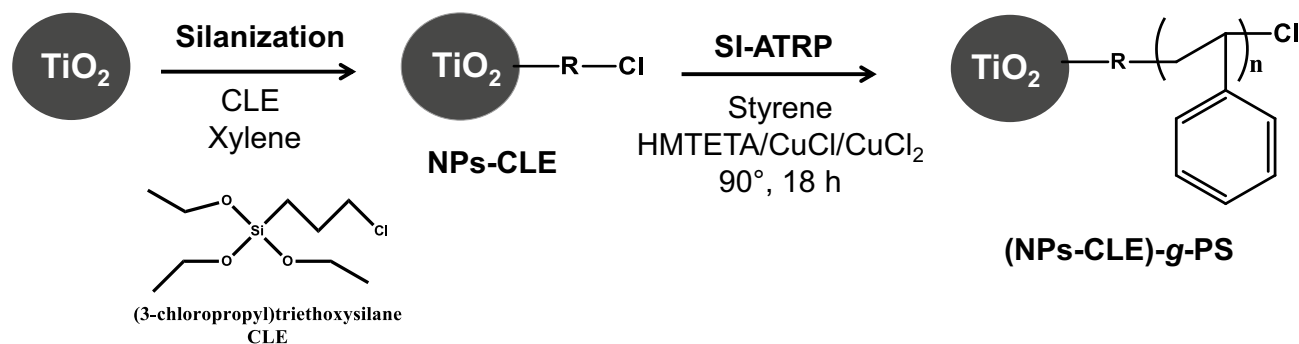


Fig. 1 DRIFT spectra of the pristine TiO₂ NPs, polystyrene, TiO₂ NPs functionalized with (3-chloropropyl) triethoxysilane (CLE) and TiO₂ NPs-CLE grafted with polystyrene via SI- ATRP. CLE/xylene ratio: 5/95

and back of droplets, respectively, just before they moved along the tilted surface. The measurements were carried out using the Drop Shape Analyzer DSA30 KRÜSS, and the contact angle was analyzed using Image J software. For each sample, three different modified PP sticks were studied, with all the measurements being repeated at least five times at different positions.

Adhesion test These measurements were performed by following a procedure previously reported by our group [18]. Concisely, a 3 M® tape was positioned in contact with the modified PP stick and then removed after 3 s. In order to observe the amount of coating removed by the 3 M® tape, images of the PP stick surface (before and after the test) and the tape were taken using an optical microscope.



Scheme 1: Core-brush hybrid materials produced via SI-ATRP on TiO₂ NPs surface with PS brushes

Results and discussion

Surface modification of TiO₂ NPs

Surface modification of the TiO₂ NPs was performed using two steps as shown in Scheme 1. First, the surface of TiO₂ NPs was functionalized by silanization process using an organosilane agent ((3-chloropropyl) triethoxysilane, CLE, which was subsequently employed in a second step, as an SI-ATRP initiator for PS grafting polymerization. Afterwards, type core-brush TiO₂ NPs were obtained with brushes of polystyrene (PS) on the surface [16, 18].

Figure 1 shows the DRIFT spectra of the unmodified TiO₂ NPs, a PS, silanized TiO₂ NPs and PS grafted TiO₂ NPs. Strong absorption bands around 3450 and 1640 cm⁻¹ were observed in bare TiO₂ NP, due to the -OH and H-O-H bending vibrations, respectively, on the surface of the NPs. In addition, a strong absorption band from 950 to 450 cm⁻¹ was observed due to the O-Ti-O stretching vibration. Upon the functionalization with CLE another band appeared at 1100 cm⁻¹. In agreement with previous studies, the new absorption band was attributed at Si-O-C stretching vibration, which confirmed a successful functionalization on the TiO₂ NP surfaces [18]. After the graft polymerization of styrene from silanized TiO₂ NPs a new strong band appeared at 1603 cm⁻¹, which was attributed to C=C vibration from the aromatic ring. Notice that the absorption band around 3450 cm⁻¹ in (NPs-CLE)-g-PS was attributable to water being strongly adsorbed at the ionic sites of the interface in the modified NPs and PS units, which was not removed during the sample preparation. These DRIFT spectroscopic results demonstrated the successful graft polymerization of styrene polymer via SI-ATRP from the functionalized surface on the TiO₂ NPs with CLE as the initiator.

The morphologies (size and shape) after surface modification of TiO₂ NPs were analyzed. When the functionalized TiO₂ NPs were grafted with PS, an apparent organic polymeric shell was observed (see Fig. 2). The TEM images revealed a different size of the shell formed by the brushes,

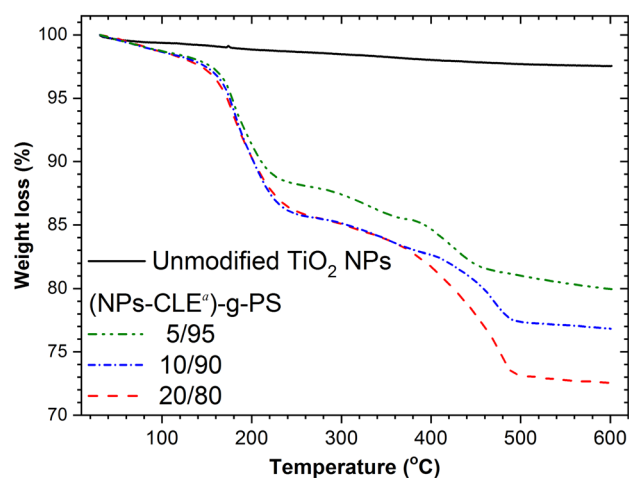


Fig. 3 TGA data of unmodified and grafted TiO₂ NPs with Polystyrene (PS) via SI-ATRP. Before PS grafting the TiO₂ NPs were functionalized with (3-chloropropyl) triethoxysilane (CLE) using three CLE/xylylene ratios, *a*: 5/95, 10/90 and 20/80

depending on the sample. The (NPs-CLE)-g-PS shell brush sizes were approximately 10.4 ± 0.5 nm, 11.5 ± 0.5 nm and 18.8 ± 0.5 nm for *a* = 5/95, 10/90 and 20/80, respectively, indicating that the polymer grafting depended on the initiator ratio. Therefore, when the ratio (organic silane/xylylene concentration) increased, the thickness of the polymer shell also increased as a consequence of controlled graft polymerization.

The thermal properties of unmodified TiO₂ NPs and PS grafted TiO₂ NPs [(NPs-CLE)-g-PS] were evaluated using TGA, as shown in Fig. 3. The weight loss of unmodified TiO₂ was less than 3% after heating from room temperature to 600 °C, which was mostly due to the evaporation of adsorbed water. For the grafted TiO₂ NPs, (NPs-CLE)-g-PS, two weight losses were observed. The first one loss (about 13%) was recorded after heating up to 230 °C, which was mostly attributable to water being strongly adsorbed at the ionic sites of the interface in the modified NPs and PS units, as previously reported [40]. All modified hybrid NPs also revealed a

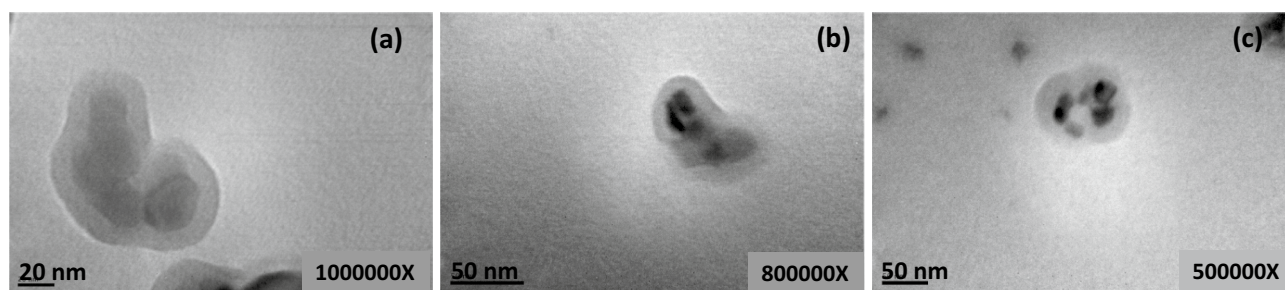


Fig. 2 TEM images of the grafted TiO₂ NPs with Polystyrene (PS) via SI-ATRP. Before PS grafting the TiO₂ NPs were functionalized with (3-chloropropyl) triethoxysilane (CLE) using three CLE/xylylene ratios, *a*: 5/95 (i), 10/90 (ii) and 20/80 (iii)

Table 2 Estimated number of surface-grafted groups per area unit, N_{exp} , and grafting percentage, %G calculated from thermogravimetric experiments

Sample	N_{exp} (nm^{-2})	%G
(NPs-CLE ^{5/95})-g-PS	101.1	8.7
(NPs-CLE ^{10/90})-g-PS	139	10.8
(NPs-CLE ^{20/80})-g-PS	208.2	16

second and main weight loss at between 230 °C and 500 °C, corresponding to a process of progressive thermal decomposition of the organic shell. This occurred in several closely related but undistinguished stages, thus making assignation of the fragments impossible. Beyond 500 °C, all thermograms remained constant. Thus, the thermal decomposition of the organic content of each system was shown to take place in the above-mentioned range of temperatures. Although a high polymerization rate of the organic silane on the surface of the TiO₂ NPs cannot be ruled out by the increase in its concentration, this effect would be noticeably less because the PS concentration is highest. Therefore, the polymeric organic content

Table 3 The gyration radius R_g and calculated radius r from unmodified NPs and PS grafted NPs, obtained by the Guiner approximation

Sample	R_g (mean $\pm$ SD) nm	r nm
(NPs-CLE ^{5/95})-g-PS	8.8 ± 0.4	129.1
(NPs-CLE ^{10/90})-g-PS	9.0 ± 0.4	135.0
(NPs-CLE ^{20/80})-g-PS	9.4 ± 0.5	147.3

of each system; as well as the organic loading achieved at every synthetic step can be calculated by taking into consideration the weight loss between 230 and 500 °C.

Clearly, a high weight loss corresponded to a high organic loading. From these weight losses, the estimated number of surface-grafted groups per area unit, N_{exp} , and grafting percentage, %G, were calculated using Eqs. 1 and 2, respectively, see Table 2. Both, N_{exp} and %G, depended on the initiator ratio, being higher when this ratio (organic silane/xylene concentration) increased. It can be observed in the %G values, 8.7%, 10.8% and 16.0% for $a = 5/95$, 10/90 and 20/80 ((NPs-CLE^{*a*})-g-PS), respectively. The same behavior was observed for N_{exp} values. The sample (NPs-CLE^{20/80})-g-PS presented the highest N_{exp} , and %G, 208.2 nm^{-2} and 16%, respectively. Notice that this TGA results showed a good correlation with the results observed in TEM images (see Fig. 2).

The gyration radius (R_g) was obtained from the SAXS measurements, see Table 3. By assuming that the scatter is due to spherical particles (a simplified assumption) and using the Guinier approximation [39], R_g was obtained from the plot of $\ln(I(q))$ versus q^2 at $q_{\text{min}} * R_g < 1.5$, with the radius of the particles (r) being derived from Eq. 3:

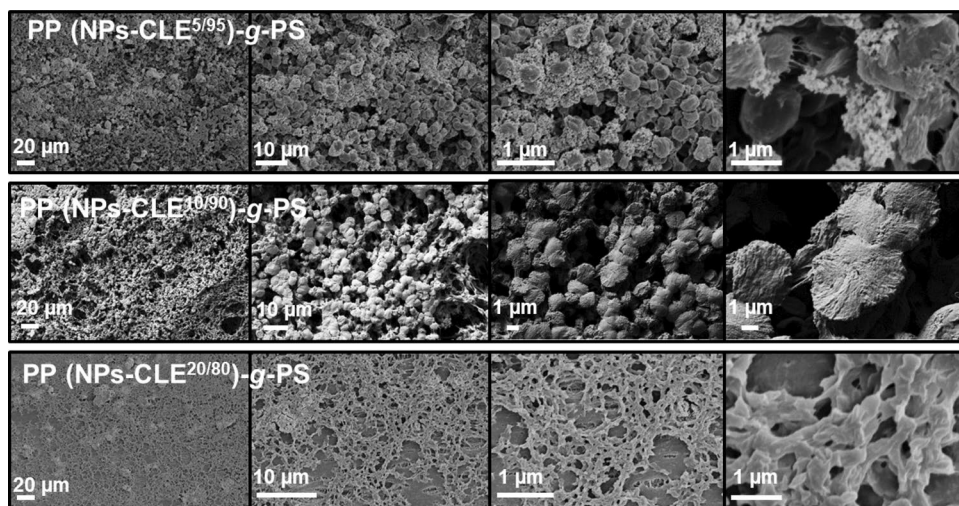
$$R_g = \sqrt{\frac{3}{5}} r \quad (3)$$

The values of r calculated for each sample were 129.1, 135.0 and 147.3 nm for the (NPs-CLE^{*a*})-g-PS with a equal to 5/95, 10/90 and 20/80, respectively. These results are in good correlation with the TEM images and TGA data, showing that polymer content depended on the initiator ratio as a consequence of a controlled graft polymerization.

Construction of superhydrophobic surfaces

The superhydrophobic surfaces were fabricated using a simple one-step dip-nano composite coating method

Fig. 4 SEM images from Polypropylene sticks after dip-coated in solutions containing grafted TiO₂ NPs with Polystyrene (PS) via SI-ATRP. Before PS grafting the TiO₂ NPs were functionalized with (3-chloropropyl) triethoxysilane (CLE) using three CLE/xylene ratios, a : 5/95 (top), 10/90 (middle) and 20/80 (bottom)



on polypropylene sticks. Briefly, NPs were redispersed in xylene at 125 °C and a PP stick was immersed for 3 s, after which, the stick was dried in an oven at 100 °C. It is important to notice that using this methodology, in which the core-brush NPs were coated on the PP surface, represents an advantage compared with other techniques for polymeric nanocomposites fabrication where NPs are distributed throughout the polymer, affecting the properties of the bulk material. Modified PP surfaces were studied using ATR-FTIR (symbol of Attenuated Total Reflectance Fourier Transform Infrared) spectroscopy, and the results demonstrated a successful surface modification of the PP sticks (Fig. S2).

The surface morphology was investigated by SEM and profilometry. Figure 4 shows SEM images of PP sticks modified with (NPs-CLE^a)-g-PS with *a* equal to 5/95, 10/90 and 20/80. In comparison with the unmodified PP sticks (Fig. S3), the modified surfaces present heterogeneous rough surfaces with hierarchical micro-nanostructures and empty spaces which is beneficial for the formation of a superhydrophobic surface [13, 41, 42].

Another important parameter for obtaining artificial superhydrophobicity is the roughness of the surface. Figure 5 presents the profilometry results of each PP stick unmodified and modified with (NPs-CLE^a)-g-PS with *a* equal to 5/95, 10/90 and 20/80. The profilometer images can be seen in the Fig. S1 (Supporting Information). The RMS roughness increased continuously from $(0.25 \pm 0.05) \mu\text{m}$ for an unmodified PP to $(1.9 \pm 0.2) \mu\text{m}$ for PP sticks modified with

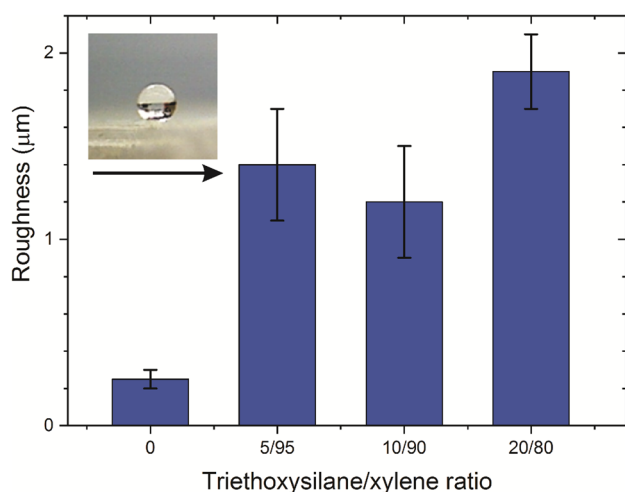


Fig. 5 Roughness data dependence of pristine polypropylene sticks and after their dip-coating in solutions containing TiO₂ NPs with Polystyrene (PS) grafted via SI-ATRP. TiO₂ NPs were previously functionalized with (3-chloropropyl) triethoxysilane (CLE) using three CLE/xylene ratios, *a*: 5/95, 10/90 and 20/80. Inset: Microscope optical image of a stick PP modified with a core-brush (NPs-CLE^{5/95})-g-PS

Table 4 Static (WCA) and hysteresis (CAH) water contact angles of PP films dip-coated in solutions containing grafted TiO₂ NPs with Polystyrene (PS) via SI-ATRP. TiO₂ NPs were previously functionalized with (3-chloropropyl) triethoxysilane (CLE) using three CLE/xylene ratios, *a*: 5/95, 10/90 and 20/80

Surface	WCA (mean ± SD)°	CAH (mean ± SD)°
Unmodified PP	110 ± 3	
PP (NPs-CLE ^{5/95})-g-PS	156 ± 4	6.2 ± 0.6
PP (NPs-CLE ^{10/90})-g-PS	160 ± 3	6.4 ± 0.6
PP (NPs-CLE ^{20/80})-g-PS	161 ± 7	3.7 ± 0.4

(NPs-CLE^{20/80})-g-PS, respectively. It can be seen that the roughness of the modified surfaces increased as the PS shell increased on the NPs.

The wettability behavior of the surfaces was investigated by measuring the static water contact angle (WCA) and the results are shown in Table 4. The WCA values increased from 110 degrees for unmodified PP to 156 ± 4 , 160 ± 3 and 161 ± 7 degrees for PP sticks modified with (NPs-CLE^a)-g-PS with *a* equal to 5/95, 10/90 and 20/80, respectively. Furthermore, the adhesion of a drop of water on the modified surfaces was studied by contact angle hysteresis (CAH). CAH is defined as the difference between the advancing and receding WCA. If the CAH is smaller than 10 degrees, the superhydrophobic surface is able to achieve the self-cleaning property by the so-called “lotus effect” [3–5], and the Cassie-Baxter state is then reached [8]. However, when CAH is higher than 10°, the superhydrophobic surface has a high water adhesion or high hysteresis (Wenzel state) [7].

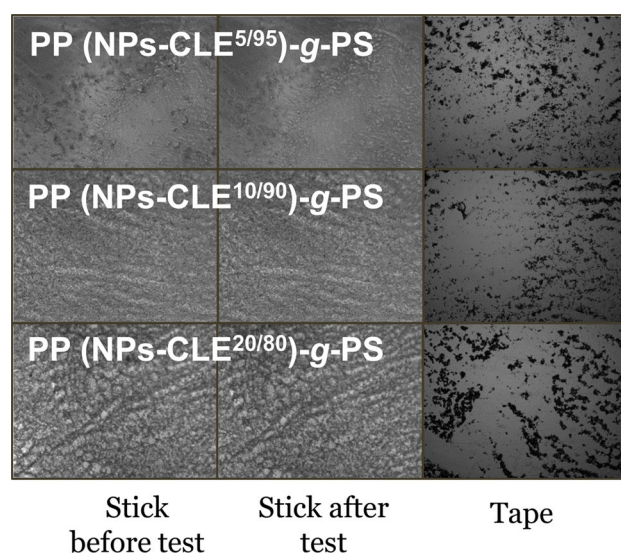


Fig. 6 Optical microscopy images of the adhesion test. For each sample, the stick before the test (left), the stick after the test (middle) and the tape after the test (right) are shown

Table 4 shows that the CAHs values obtained for the PP surfaces modified with (NPs-CLE)-g-PS are smaller than 10 degrees, which indicates a very low hysteresis and self-cleaning conditions for all the modified samples. Figure 5 (inset) shows an image from a PP stick modified with core-brush (NPs-CLE^{5/95})-g-PS obtained by optical microscopy, where the superhydrophobic property of the surface is evident (see [video material](#) in Supporting information). It is important to notice that the wettability properties on PP modified surfaces (WCA and CAH) remained constant for at least one year.

The coating adhesion to the PP stick was evaluated by a simple test using 3 M® tape [18]. Figure 6 shows three optical microscope images obtained for each sample, which corresponded to the modified PP stick before the test (left), the modified PP stick after the test (middle) and the tape after the test (right). In spite the test is qualitative, it is possible to infer that the adhesion properties were improved in comparison with our previously described superhydrophobic surfaces [16, 18]. In all the samples evaluated, the tape images presented a minimum amount of coating removed. Moreover, the images of the modified PP stick revealed that its morphology was unchanged after the test, which can be attributed to a covalent bond of the hydrophobic polymer (PS) on the NPs surfaces.

Conclusions

Durable superhydrophobic PP surfaces with super-repellent and self-cleaning properties were successfully prepared by a facile method which involves PP dip-coating using core-brush TiO₂ NPs dispersed into xylene. Core-brush TiO₂ NPs with PS brushes were produced by in situ graft polymerization using SI-ATRP. The PS grafting on the TiO₂ NPs surfaces was confirmed by the DRIFT, TGA, TEM and SAXS data; it was observed that the polymer content on the TiO₂ NPs depended on the initiator ratio, as a consequence of a controlled graft polymerization mechanism. Furthermore, the dip-coated PP substrates revealed a surface morphology by SEM with hierarchical micro-nanostructures and empty spaces, as well as higher RMS compared with pure PP, which was beneficial for attaining a superhydrophobic surface with low hysteresis.

In the present study, the results showed that the use of styrene as a monomer to graft polymer chains on the surface of TiO₂ NPs was an efficient and low cost choice for the preparation of superhydrophobic surfaces. The interactions between chains may occur by styrene rings that reduced the chain mobility allowing a decrease in the surface energy of the material. Finally, it led to the superhydrophobicity self-cleaning properties observed.

The present investigation provides an alternative procedure to prepare polymer superhydrophobic surfaces by adapting the preparation of core-brush NPs and micro-nanostructures formed on the thermoplastic polymer surface and it could be extended to different surfaces. Finally, due to the easily controlled superhydrophobic properties and excellent adhesion of the coating, it is expected that this procedure can improve the technological application in water repellent polymer substrates.

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